

4 (short)

(a) For the inverting op-amp circuit of Fig. 5, and assuming the op-amp is ideal, derive an expression for the voltage gain in terms of R_1 and R_2 .

[3]

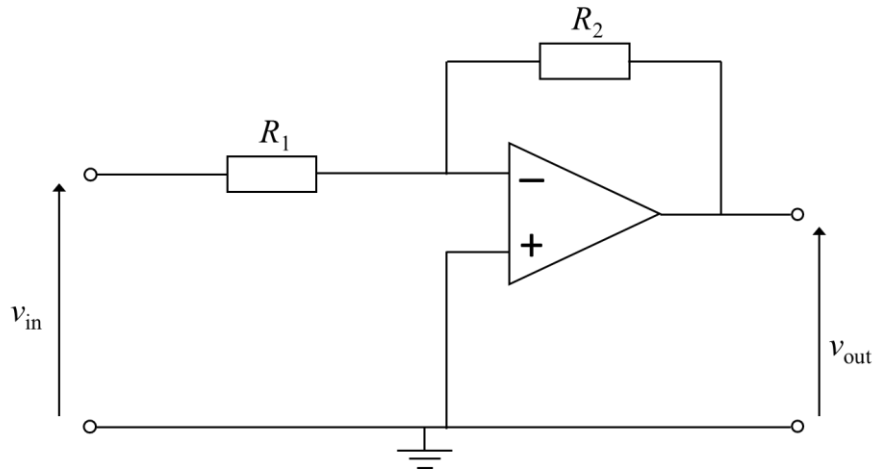


Fig. 5

(b) Figure 6 shows the circuit diagram of a difference amplifier. Assuming the op-amp is ideal, calculate the voltage gain $\frac{v_{out}}{v_2 - v_1}$.

[7]

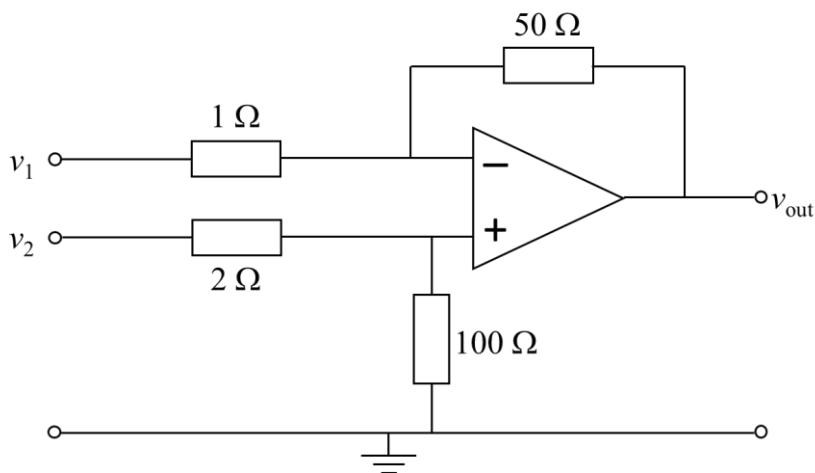


Fig. 6